

Modeling Highway Functionality of Automated Vehicles Using Finite Automata

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Introduction:

In our modern day and age, the production and widespread consumer use of automated vehicles is a rapidly expanding field. Safety concerns, a legitimate deal-breaker for any user, raise doubts about the reliability of autonomous vehicles. These questions of safety and concern regarding proper functionality must be addressed and solutions proposed in order to enhance growth in this revolutionary and exciting area of technology. In this paper, our literature review will break down models which several other researchers have proposed as solutions to the functionality concerns of these automated vehicles and our implementation step will propose our own finite automata model as a solution to these concerns.

For our research purposes, we will review literature which focuses mainly on specific circumstances of three-lane highway driving, since we do not have enough room in this paper to discuss more specialized scenarios and since highway driving often results in a wide variety of states for a vehicle. Our sources also include specific algorithms for synthesizing data from controlled experiments to improve safety through repeated trials. Throughout our paper, we used the term “automated vehicles” to describe vehicles which use technology to assist or remove the driver. Building off of this peer-reviewed work, we will then construct a finite automata which can effectively model a subset of the many functionalities and real-world scenarios which an automated vehicle could face while driving on a three-lane highway.

Our immediate goal in creating this finite model is to thoroughly engage and grapple with how these kinds of models can represent real-world states and actions. Once we have grasped these fundamental connections, we can further understand patterns which can be modeled using these finite state acceptors. Therefore, the ultimate goal and purpose of this research is to efficiently model the *safest* and *most optimized* automated vehicle that we can

using the tools we have been given. Thus, the central inquiry of this paper will be: How can finite automata theory be used to model the basic functions of automated vehicles and their “decision-making” output, particularly with regards to driving in traffic flow, to make it a safer, more reliable means of transportation?

Literature Review & Analysis:

The first theme in our literature which stood out as an effective way to begin implementing finite automata theory for automated vehicles was the modeling of safe lane changes. Our first two sources explore this possibility using finite state machines to accomplish these actions efficiently and safely. In their article, Jaswanth et al. determine that a finite state machine is a fitting tool to describe functions such as adaptive cruise control, lane-keeping, and autonomous emergency braking. As seen in Figure 2, they elect to use JFLAP (the same Java-based tool which we will use) to design the non-deterministic finite acceptor for their research. (Jaswanth et al., 2022) Their implementation uses the form $M_1 = (Q, \Sigma, \delta, q_0, F)$ to describe 14 possible states including an initial and final state (q_0 and F) and 28 transitions (δ) using the an alphabet of 25 actions (Σ) which shift between various states (Q). (Jaswanth et al., 2022) They define the sensing system of the automated vehicle to determine the lengthwise distances between the cars in proximity to the vehicle on the highway. This sensing system surveys eight different directions at increments of 45 degrees, accounting for full 360 degree sensing capability.

An example scenario, demonstrated in Figure 1, depicts the automated car stuck in the first lane in a traffic scenario, trapped by the car in front of it and to its left. Here, the results are switched to a mirror image since the highways outside of America have flipped lane precedence, so our reviews are adjusted accordingly. The finite state machine takes in the different states

which the car is in and, using the sensing system defined earlier, switches states based on the movement of the proximate cars. If the car ahead accelerates, our vehicle would accelerate as well. Now that our left is clear, assuming the car to the left doesn't accelerate and there are no other cars in the middle lane, our left sensors would indicate that we can safely shift into a new state. (Jaswanth et al., 2022)

A second model proposed by Bae et al. (2020) describes the flow of a vehicle's autonomous driving capabilities using three criteria: perception, planning, and control. They describe perception as the ability to take in information about the driving environment, similar to the sensing system described by Jaswanth et al. (2022) in their article. The planning criteria refers to the description of actions which constitute the machine's state transitions or reactions to the perceived environment. Finally, the control step executes these state transitions using the appropriate algorithms and control sequences. (Bae et al., 2020) Whereas Jaswanth et al. (2022) were a bit vague in their description of the mode of the vehicle's sensing, Bae et al. (2020) have a far more in-depth explanation, shown in Figures 6-7, of how the sensors actually capture images, track objects, and recognize obstructions using Lidar, Camera, Global Navigation Satellite Systems, and Tachometer technology. These technologies extract lane width, center, curvature and heading angle and send this data to be processed based on the resultant planning and control flow. (Bae et al., 2020) Next, a flow chart is created to plan out the necessary state transitions for our vehicle. Then, the authors create two finite state machines to model the control (acceleration, deceleration, stopping, etc.) and motion of the vehicle (lane shifting, staying in lane, etc.). The motion finite state machine can account for turning as well as lane changing, a factor which Jaswanth et al. (2022) did not account for in their model, but which could be a useful factor to consider when trying to accommodate curved highways. Upon reviewing and

synthesizing these two articles, we found that Bae et al. (2020) provided a more in-depth modeling of automated vehicle behaviors. Not only did they incorporate an explanation for how the sensing systems operated, but they also set forth a well-structured, three-step model for how a vehicle could go about taking in input data from these sensors and responding appropriately. Altogether, these two articles present helpful connections to our theme of safe lane changes and lay out the basic functionality and steps needed to perform these actions using finite state machines.

The second theme which we will explore stems from three sources which connect the need for modeling human-based interactions and behaviors when designing automated vehicles. In their article, Ransikarbum et. al (2017) propose a design system for highway driving using Affordance-Based Finite State Automata (ASFA) modeling. The purpose of ASFA modeling “incorporates dynamic and perceivable properties of affordance into a formal control model in such a manner that a human operator has a set of possible actions and can take an action based on perceived systems.” (Ransikarbum et. al, 2017) This type of model allows the vehicle’s finite automata model to specifically transition into other states based on certain human driver behaviors and interactions. The literature review presented below demonstrates the safety measures each author takes to illustrate the widespread future utility of autonomous vehicles.

The authors develop an ASFA model based on a Highway-Lane-Driver-System (HLDS) problem, depicted in Figure 4, in which there are three lanes, two drivers, and one exit. In this scenario, driver one is trying to take the exit (Ransikarbum et. al, 2017). Using the ASFA model, driver one’s transition to different states, corresponding to the different lanes, depends on where the second driver is, emphasizing its goal of safety to prevent collision with the second driver in order to successfully and safely reach its final state, the exit. To further test the ASFA model and

its effectiveness, Rothrock et. al (2017), implement the same HLDS model on a real driving track with a subject driver and confederate driver. The confederate driver's path was pre-determined so that the experiment can solely focus on the subject driver's behavior. The goal of this experiment was to analyze the subject driver's behavior and lane changes, influenced by the ASFA system's transitions, until the exit was reached. The authors state that "the statistical results show that the agent-based AFSA simulation fits well with the driver's behavior in the designed experiment for 94% of all the trials" (Ransikarbum, et al., 2017). These successful results validate the applicability of finite automata theory to this proposed model, suggesting the potential of the ASFA modeling of human interactions for automated vehicle design. Not only this, they also connect back to the first theme in our review and describe a real-world experiment to gather data on how to replicate human behaviors when performing a lane shift or cut-in.

Hwang et al. (2022) take a different approach, seen in Figure 5, in creating a finite state machine for their self-driving car model. They incorporate reinforcement learning into their finite state machine in order to ensure "performance in a variety of risky situations" and to guarantee safety using their finite state machine (Hwang et al., 2022). Similar to the experiments performed by Ransikarbum et. al, the authors focus on the safe lane-merging aspect of highway driving. The cut-in algorithm used to test their model operates as follows: An initial phase titled "ready" transitions into a next phase titled "approach" which begins the attempt at a lane change. In the approach phase, the algorithm will access the gap risk evaluation and determine if it is safe or risky. If it is safe, the model will transition into a lane change phase where "the policy-based Reinforcement-Learning (RL) algorithm executes a lane-change maneuver by controlling the acceleration and steering angle of the ego vehicle" (Hwang et al., 2022). Otherwise, it will try to "negotiate" with the other driver in order for the model to try a cut-in maneuver. The RL model

can control the acceleration and steering angle to transition into another state in order to reduce stress on the finite state machine. Since the RL model cannot ensure safety on its own, the finite state machine “guarantees safety [and] performance can be maximized by using a learning algorithm in the lane-changing phase” (Hwang et al., 2022).

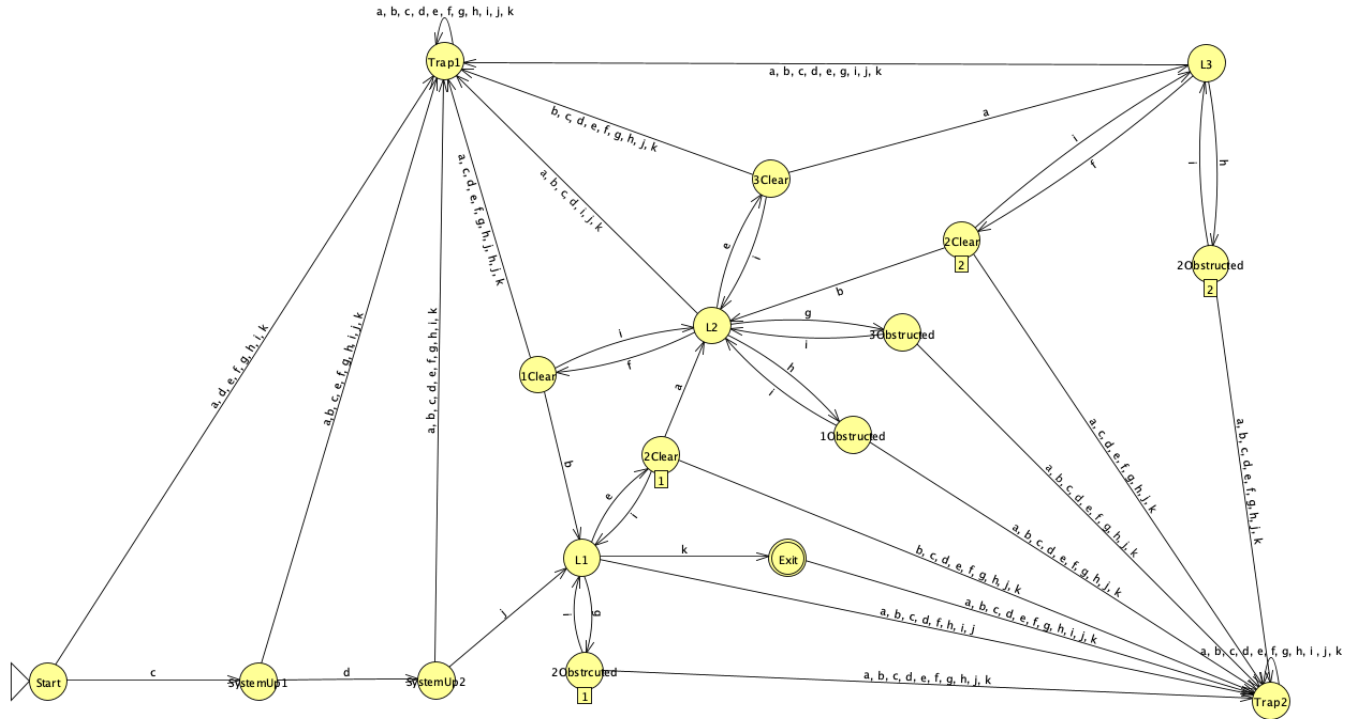
The authors were able to simulate and have success with this model by “consistently achiev[ing] a high cut-in success rate without sacrificing safety.” The FSM-RL model demonstrates a high safety and success rate in lane merging and cutting, demonstrating its reliability and the possibility of its future implementation in autonomous vehicles. The model’s decision making based on the RL approach demonstrates the safety and reliability of the model and its means for transportation. This RL model connects to our theme of simulating human response in driving situations as a response to the roadway environment as well as drawing from our first theme of lane changing using a state transition machine, demonstrating the great utility of this model in understanding automated vehicle design. Capturing and envisioning how a human can learn the different scenarios experienced when driving is incredibly important in realizing the possibility of this kind of state machine operation used as a design tool. Similar to a student in a driver's education course, the RL-FSA model could take in and store the many different driving situations (i.e. lane-merging, tight turns, exits, avoiding obstacles, etc.), efficiently and safely learning from its environment through a repeated data input feed.

In the final article, Bouthoute et al. (2014) develop a hybrid i/o automata based on a system called VANET, which extracts information based on human driver behaviors. The use of the VANET model ensures the safest possible transitions while on the road. The model’s goal is to analyze the human drivers’ behavior according to the following three categories: Convenient, Tolerable, and Risky behavior. Analyzing these behaviors will allow the model to make

decisions and perform state transitions based on the information received from their surroundings (Bouthoute et al., 2014). Unlike Raniskarbum et al. (2017), who propose that the decisions made are contingent upon the second driver's actions, Bouthoute et al. (2014) propose gathering more data based on the driving environment so that the model can safely handle any transition. The hybrid i/o model, glimpsed partially in Figure 8, significantly emphasizes the safety precautions it takes before transitioning into another state to prioritize its reliability in designing this means of transportation. The VANET model shares similarities with the testing models proposed by the other authors, and helps to visualize the modeling of human behavior in the driving process.

These three sources help to give a good picture of how a state transition graph could be used to model slightly less predictable human behaviors when driving, leaving us with some questions as to how we could simply implement this idea into our own DFA. The algorithms used in some of these studies are fascinating due to their efforts to construct a machine using pattern recognition and data synthesis describing human behavior, but how could we represent this in our own model? As described later in our implementation section, we attempted to model some human “decision-making” behavior in our DFA. The way we chose to apply this behavior is, when faced with a successful lane scan, to enter a state described as 1Clear, 2Clear, or 3Clear depending on the lane we are transitioning to. We resolved our initial middle lane overlap issues by splitting up the 2Clear states into 2Clear(1) and 2Clear(2), distinguishing clear from the left side from clear from the right side. Our model used our findings from the second theme's sources to make possible our vehicles ability to successfully scan to the left or the right, yet still stay in the lane, modeling a human decision in the driving process. Whereas a regular state transition graph would not have the flexibility to make this “choice”, we implemented this human-based decision into our finite state machine based on our thematic analysis and literature review.

Our Proposed Finite Automata Implementation:



We constructed our DFA called AV_1 using the Java-based software tool, JFLAP. The DFA follows the format $M_1 = (Q, \Sigma, \delta, q_0, F)$ and the states and edges are defined as follows:

Q:

- Start** - System starting up
- SystemUp1** - Preliminary state for Highway Cruise System Activation
- SystemUp2** - Preliminary state for i/o Braking & Yield System Activation
- L1** - In Lane 1, **L2** - In Lane 2, **L3** - In Lane 3
- 1Clear** - Lane 1 was scanned and is clear of obstructions
- 2Clear (1)** - Lane 2 was scanned from lane 1 and is clear of obstructions
- 2Clear (2)** - Lane 2 was scanned from lane 3 and is clear of obstructions
- 3Clear** - Lane 3 was scanned and is clear of obstructions
- 1Obstructed** - Lane 1 was scanned and there is a vehicle or object obstructing it
- 2Obstructed (1)** - Lane 2 was scanned from lane 1 and there is a vehicle or object obstruction
- 2Obstructed (2)** - Lane 2 was scanned from lane 3 and there is a vehicle or object obstruction
- 3Obstructed** - Lane 3 was scanned and there is a vehicle or object obstructing it
- Trap1 & Trap2** - Account for all unused transitions for each state
- Exit** - Vehicle takes an exit off of lane 1

Σ :

- a.** Shift Left
- b.** Shift Right
- c.** Activate Highway Cruise-Control System
- d.** Activate i/o Braking & Yield System

- e. Successful Left Lidar Scan
- f. Successful Right Lidar Scan
- g. Unsuccessful Left Lidar Scan
- h. Unsuccessful Right Lidar Scan
 - i. Stay in Current Lane
 - j. Activate Lidar Scanner
 - k. Take Exit

Design Details:

Taking inspiration from the designs of Jaswanth et al. (2022) as well as Bae et al. (2020), we developed a deterministic finite acceptor to model simple highway driving functionality of an automated vehicle. We parsed together bits and pieces that we liked and disliked from their work and added our own ideas, culminating in our own new model. Our implementation uses a Lidar and Camera Scanner similar to the one proposed by Bae et al. (2020) but with simplified scanning abilities. Rather than increments of 45 degrees, our DFA can only handle 180 degree scans to the left and right, a limitation of our design. To account for this, we developed the idea of beginning every accepted string in our language by activating three systems in our vehicle, a kind of sequential booting up process: The Highway Cruise-Control System assisted by Global Navigation Satellite Systems (GNSS), The i/o Braking & Yield System, and the Lidar Scanner System to check for obstructions. It is not possible to account for every possible state which a vehicle could encounter in such a small amount of space, but by implementing these three systems, it gives the reader an idea of how edge cases could be handled.

For example, since our DFA cannot yet account for another vehicle or object directly in front of our vehicle and respond by braking hard or slowing down quickly, we must resort to a more general system which we simply called the Braking & Yield System. This system responds to scanner input and involves taking in data based on the input. The Highway-Cruise Control System is based on GNSS data which takes into account which highway the vehicle is located on

and adjusts to the speed limit accordingly, by automatically controlling the engine's RPM using tachometer technology. This is necessary in order to keep our automated vehicle up to speed with the flow of traffic and the GNSS systems also helps to identify potential traffic or road obstructions up ahead, inevitably leading to use of the aforementioned Braking & Yield System based on the given data. Though these three systems are not yet fully implemented into our DFA design, there is certainly room for future improvements in this area so that more situational input can be properly processed.

Results Explaining a Sample String Accepted by our Model - “cdjgieieaeafbfbk”

The sequence of real-world actions which this string represents is an automated vehicle shifting through all the lanes of a highway and back. The vehicle travels in the middle lane for most of the journey, hops into the fast lane, and finally shifts back over two lanes and exits the highway safely. The first three symbols (c, d, and j) represent the three systems starting up in our vehicle, a necessary prefix for any accepted string in our language. Our Lidar Scanner is always actively scanning and searching for obstructions to the left and right, allowing us to always check whether or not we are clear to switch lanes. Even if the lane is clear, our DFA allows for our vehicle to remain in the same lane, which it elects to do in the sequence “giei”. Next the vehicle, having detected that the coast is clear and that we can now shift into the middle lane, shifts left. Checking again whether or not the coast is clear it shifts left once more. (“eaea”) Now, we are in Lane 3 (the fast lane). The scanner checks for right obstructions in Lane 2, finds none, and shifts right. The same sequence happens again to get our vehicle into Lane 1. (“fbfb”) Finally, having completed our shifts, the vehicle exits the highway and our string is complete. (“k”)

Conclusion:

In conclusion, we made a great deal of progress in understanding the nuances of automated vehicle design. We loved the idea of implementing light detection and ranging (LiDAR) as seen in Figures 6-7, a technology which was previously used in topography, into our own design. Although we did not go into all the specifics of how this technology operates, Bae et al. (2020) clearly define its functionality, according to our purposes, in their article. As was described earlier, we faced some challenges due to transition overlap when attempting to shift back into the middle lane. This was resolved by a separate state to check for a successful scan of the middle lane from both the left and the right lanes. There are certainly many areas in which we could improve our design, for example, implementing a turning action to control any curves in the highway, adding full 360 degree scanning capabilities in order to account for emergency braking and object avoidance, and describing how a vehicle which needed to get into lane 1 in order to exit would safely adjust its speed and steering angle in order to safely perform the necessary cut-in operations. Overall, we learned a great deal about the use of finite state machines in the field of automated driving, exploring and brainstorming the many different states and actions which a vehicle of this nature could take in order to get from point A to point B most safely and most efficiently.

Appendix

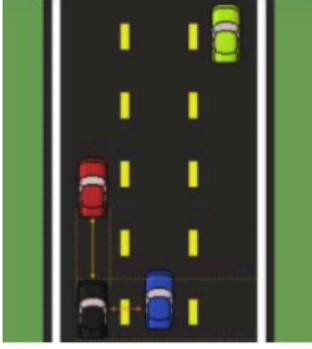


Figure 1: Jaswanth et al. (2022)

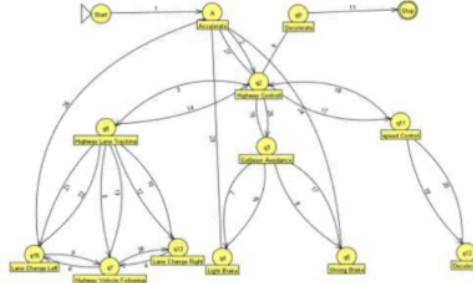


Figure 2: Jaswanth et al. (2022)

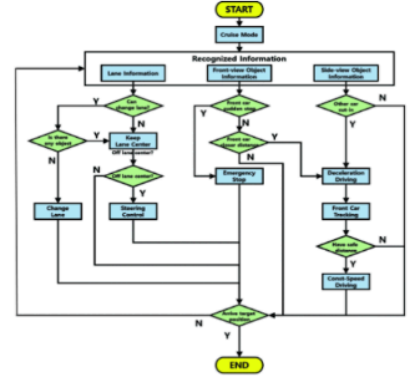


Figure 3: Bae et al. (2020)

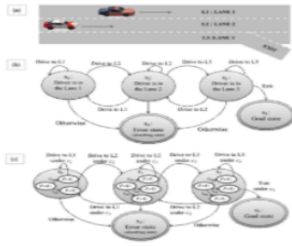


Figure 4: Ransikarbum et al. (2017)

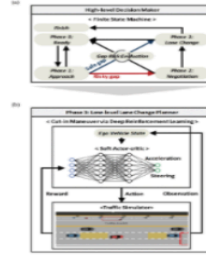


Figure 5: Hwang et al. (2022)

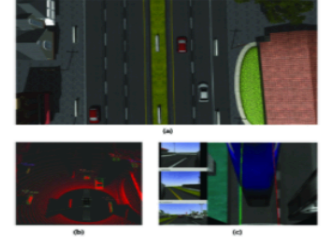


Figure 6: Bae et al. (2020)

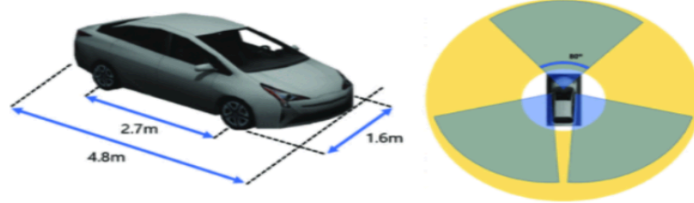


Figure 7: Bae et al. (2020)

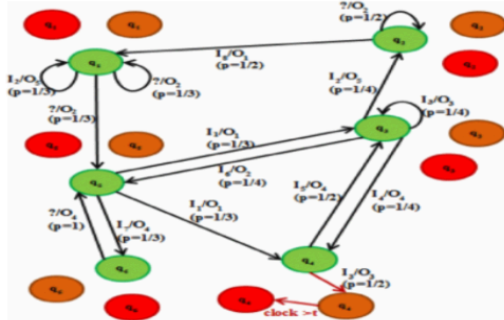


Figure 8: Bouhoute et al. (2014)

Received input		Driver's output	
Action	Variable		
-	-	O_2 : Accelerate	
I_2 : give right of way	$speed_{min} = 0$ $speed_{max} = 20$	O_2 : Give way	
-	-	O_2 : Accelerate	
I_1 : allowed speed 70kmh	$speed_{max} = 70$	O_1 : Decelerate	
I_3 : No overtaking	$lane_{min} = 0$ $lane_{max} = 0$	O_2 : Keep lane	
I_4 : Road construction	$lane_{min} = 1$ $lane_{max} = 1$	O_2 : Change lane	
I_5 : End of road construction	$lane_{min} = 0$ $lane_{max} = 0$	O_2 : Change lane	
I_6 : End of allowed speed	$speed_{min} = 0$ $speed_{max} = 120$	O_2 : Accelerate	
I_7 : Accident warning	$lane_{min} = 0$ $lane_{max} = 0$ $\theta_{min} = 5$	O_2 : Change lane	
-	-	O_2 : Change lane	
I_1 : allowed speed 70kmh	$speed_{max} = 70$	O_1 : Decelerate	
-	-	O_2 : Change lane	
I_3 : No overtaking	$lane_{min} = 0$ $lane_{max} = 0$	O_2 : Keep lane	
I_2 : give right of way	$speed_{min} = 0$ $speed_{max} = 20$	O_2 : Give way	
-	-	O_2 : Accelerate	
I_8 : Parking	$speed_{min} = 0$ $speed_{max} = 20$	O_1 : Decelerate	

Figure 8a: Bouhoute et al. (2014)

State ID	Description			
	V	lane	θ	d
q_1	[0,30]	0	[0,15]	$> V/2$
q_2	[30,50]	0	[0,15]	$> V/2$
q_3	[50,70]	0	[0,15]	$> V/2$
q_4	[50,70]	1	[0,15]	$> V/2$
q_5	[70,90]	0	[0,15]	$> V/2$
q_6	[70,90]	1	[0,15]	$> V/2$

Figure 8b: Bouhoute et al. (2014)

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